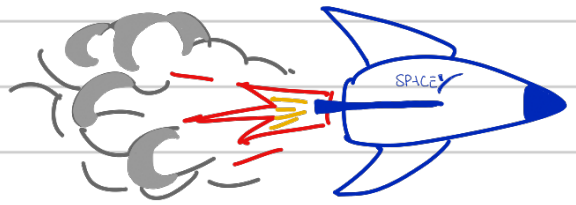
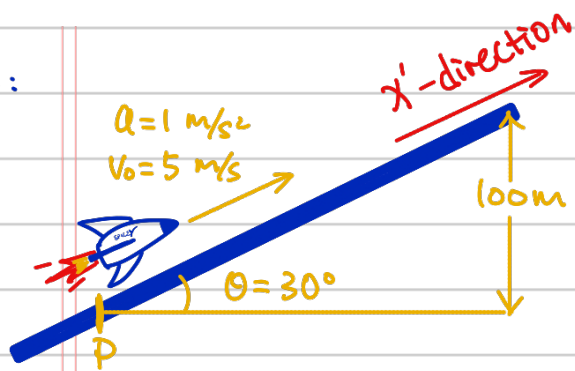


Setup:



Step 1.

the rocket is travelling on the incline (hypotenuse)
We want it to go up by 100 m (opposite side)
→ trigonometry!!

$$\sin(30^\circ) = \frac{100 \text{ m}}{\Delta x'}$$

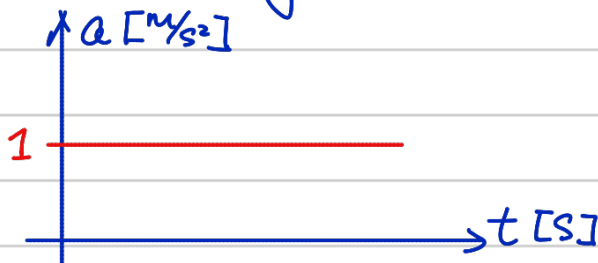
$$\sin(30^\circ) = \frac{1}{2}$$

$\Delta x'$ is displacement on the incline

$$\text{so } \boxed{\Delta x' = 200 \text{ m}}$$

Step 2

acceleration is kept constant $a = 1 \text{ m/s}^2$ over time
→ flat straight line



Step 3

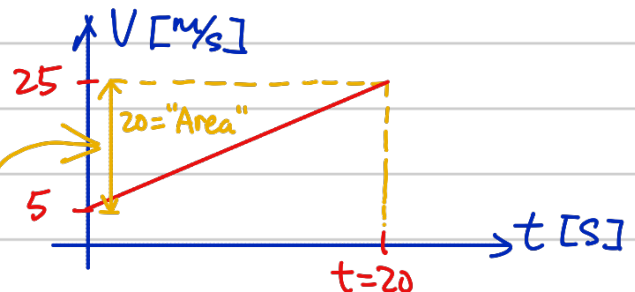
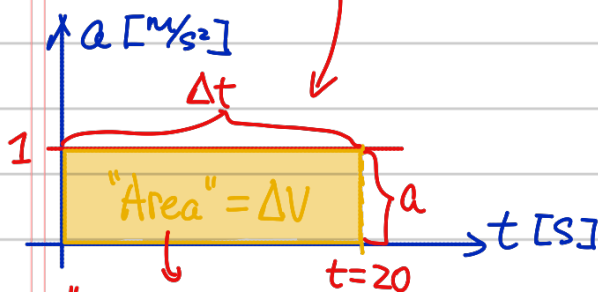
remember $a = \lim_{\Delta t \rightarrow 0} \frac{\Delta V}{\Delta t}$ if a is constant, ignore " $\lim_{\Delta t \rightarrow 0}$ "
→ $\Delta V = a \cdot \Delta t$ for constant acceleration.

$$\text{Change in velocity } \Delta V = V_{\text{final}} - V_{\text{initial}}$$

you pick a final time and this is instantaneous velocity at that time. In this case we pick $t_{\text{final}} = 20 \text{ s}$

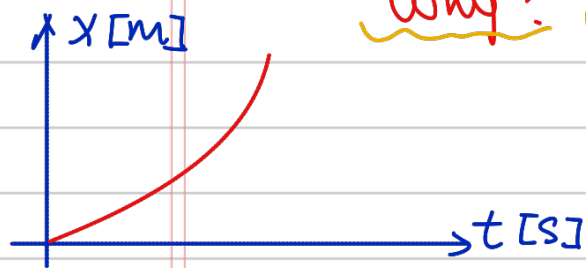
this is $v_0 = 5 \text{ m/s}$ at $t = 0 \text{ s}$ (given in the problem)

$$\text{so } \Delta V = a \cdot \Delta t = (1 \text{ m/s}) \cdot (20 \text{ s} - 0 \text{ s}) = 20 \text{ m/s}$$



Step 4

As you have seen many times (Week-1 worksheet) when velocity is a tilted straight line, position curve should be a **parabola**, more specifically, v is growing in this problem, so the parabola is **concave-up**. Since v is always larger than 0 in this problem, we only have the **right-half of the parabola**.



Why?

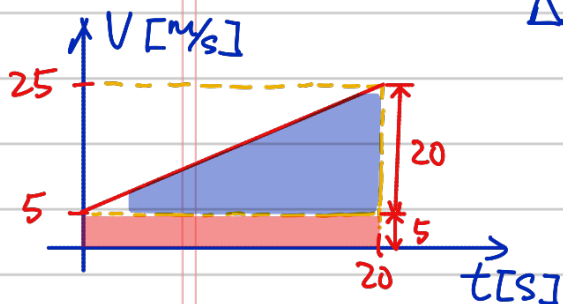
because rocket is moving faster and faster to the right
→ x grows faster and faster over time, never decreases

Step 5

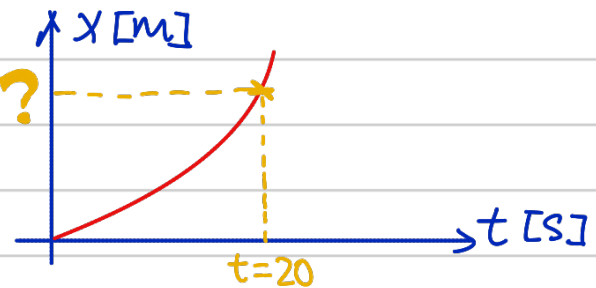
to do this, recall what we have

$$\Delta x'_{0 \rightarrow 20} = \text{"Area" under } v \text{ from } 0 \text{ s to } 20 \text{ s}$$

displacement from $t=0$ to $t=20$ s
(or change in position)



we want to find this number → ?



$$\Delta x'_{0 \rightarrow 20} = \text{"Area" under } v \text{ from } 0 \text{ s to } 20 \text{ s}$$

$$= \text{Area of } \triangle + \text{Area of } \square$$

$$= (20 \text{ s}) \cdot (25 \text{ m/s} - 5 \text{ m/s}) / 2 + (20 \text{ s}) \cdot (5 \text{ m/s})$$

$$= 200 \text{ m} + 100 \text{ m} = \boxed{300 \text{ m}}$$

so the rocket has traveled 300 m in 20 seconds, which is more than the 200 m ^{step 1} that we want for it to go up 100 m vertically.
→ it takes less than 20 s for it to travel 200 m on the ramp

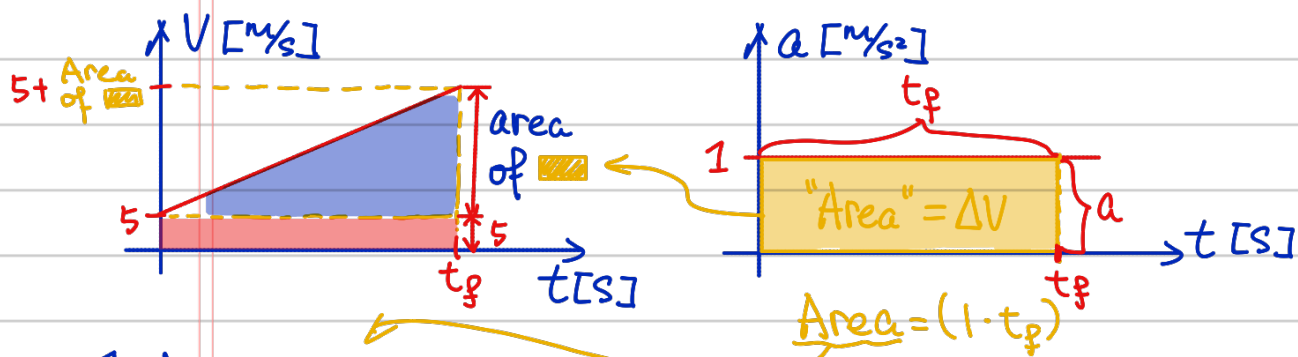
Step 6

For it to travel 200 m on the ramp. set $\Delta x'_{0 \rightarrow ?} = 200 \text{ m}$

$$\Delta x'_{0 \rightarrow ?} = \text{"Area" under } v \text{ from } 0 \text{ s to } ? \text{ s} = 200 \text{ m}$$

$$= \text{Area of } \triangle + \text{Area of } \square$$

(see pictures on next page)



Area = $(1 \cdot t_f) \cdot t_f / 2$

Area = $5 \cdot t_f$

$\Delta x'_{0 \rightarrow t_f} = \text{"Area" under } V \text{ from } 0 \text{ s to } t_f \text{ s} = 200 \text{ m}$
 $= \text{Area of } \triangle + \text{Area of } \square$

$200 = (1 \cdot t_f) \cdot t_f / 2 + 5 \cdot t_f$

quadratic equation for t_f

Solve for t_f using quadratic formula

$t_f = \frac{5^2 \pm \sqrt{5^2 - 4(\frac{1}{2})(-200)}}{2 \cdot (\frac{1}{2})} = 15.6 \text{ s}$

keep only positive t

time it takes for it to go up 100 m vertically.

Step 7

$200 = (1 \cdot t) \cdot t / 2 + 5 \cdot t \leftarrow \text{eqn we get in step 6}$
 $x - x_0 = a t^2 / 2 + v_0 t \leftarrow \text{formula in step 7}$

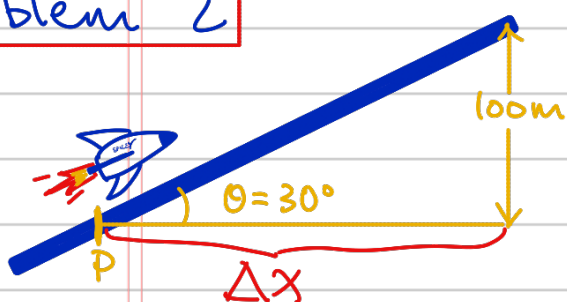
We have just solved for this quadratic eqn. $t = 15.6 \text{ s}$.

* This is where the formula comes from.

* In the future, if you see similar problem again with constant acceleration, apply the formula directly, no need to calculate the "areas" again.

Problem 2

trigonometry !!



$\tan(30^\circ) = \frac{100 \text{ m}}{\Delta x}$

$\Rightarrow \Delta x = \frac{100 \text{ m}}{\tan(30^\circ)} = 100\sqrt{3} \text{ m}$